

The Hidden Quotient

Qualified theorem, worked examples, and a corrected geometry translation

Corrected investigation 1.1 8 September 2026

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This edition corrects the supplied nine-page *Hidden Quotient Investigation*. It keeps its three-example structure, restores the finite-measure qualification already present in the project's v1.7, and separates information loss from failure to represent a desired object. The investigation's edition number does not rename the v1.7 mathematical source.

The core packages established results about multiplication operators and localizable measure spaces. The project name is an expository label. The geometry translations below have their own declared domains, observations, and witnesses.

Reading route. Pages 1–3 give the kernel, reduction, structural theorem, and qualified Radon–Nikodym statement. Pages 4–5 contain the three examples. Pages 6–8 give the corrected translation and a fully specified dynamics example. Page 9 records repairs and sources.

1. The fixed multiplication witness

Start with actual bounded measurable functions:

$$\mathcal{B}(X, \Sigma) = \{f : X \rightarrow \mathbb{C} : f \text{ is measurable and bounded}\}.$$

For a measure space (X, Σ, μ) , multiplication defines

$$\pi : \mathcal{B}(X, \Sigma) \longrightarrow B(L^2(X, \mu)), \quad \pi(f) = M_f, \quad M_f g = fg.$$

This is a contractive $*$ -homomorphism, preserving the identity operator. Its kernel depends on the declared L^2 representation.

A measurable set A is *locally null* if $\mu(A \cap E) = 0$ for every measurable E with $\mu(E) < \infty$. Write $\mathcal{N}_{\text{loc}}$ for this ideal of sets. A function vanishes locally almost everywhere if its restriction to every such E vanishes almost everywhere.

Kernel formula.

$$\ker \pi = \{f \in \mathcal{B}(X, \Sigma) : f = 0 \text{ locally a.e.}\}.$$

Proof. If $M_f = 0$, test against $\chi_E \in L^2$ for each finite-measure E . Conversely, if $g \in L^2$, the sets $E_n = \{|g| \geq 1/n\}$ have finite measure by Chebyshev's inequality and cover $\{g \neq 0\}$. Local vanishing of f gives $fg = 0$ almost everywhere. $\square$

Thus the operator can discard more functions than ordinary almost-everywhere identification does. The extra loss disappears precisely when μ is semifinite.

2. Reduction and the structural theorem

Define the semifinite reduction on the same measurable space by

$$\mu_{\text{sf}}(E) = \sup\{\mu(F) : F \in \Sigma, F \subseteq E, \mu(F) < \infty\}.$$

Then $\mu_{\text{sf}} \leq \mu$, the measures agree on finite- μ sets, and μ_{sf} is semifinite. Also $\mu = \mu_{\text{sf}}$ exactly when μ is semifinite, and μ_{sf} -null sets are precisely $\mathcal{N}_{\text{loc}}$. The L^1 and L^2 spaces for the two measures are canonically isometrically identifiable.

The quotient and operator norm are

$$\mathcal{B}(X, \Sigma) / \ker \pi \cong L^\infty(\mu_{\text{sf}}), \quad \|M_f\| = \|f\|_{\infty, \text{loc}},$$

$$\|f\|_{\infty, \text{loc}} = \inf\{c \geq 0 : \{|f| > c\} \in \mathcal{N}_{\text{loc}}\}.$$

The quotient map into operators is faithful and therefore isometric as a $*$ -representation. The usual μ -essential supremum equals this norm for all bounded measurable f exactly when μ is semifinite.

Two operations. Semifinite reduction changes the measure and its null ideal; it does not supply missing suprema in the quotient Boolean algebra. Localizable completion addresses those missing suprema. It may require a larger algebra or a different measure-space realization. These are distinct constructions.

Use *localizable* in Fremlin's sense: semifinite with Dedekind complete measure algebra. Since μ_{sf} is already semifinite, its localizability means that $\Sigma/\mathcal{N}_{\text{loc}}$ is Dedekind complete.

Structural theorem. The following are equivalent:

1. μ_{sf} is localizable.
2. The canonical isometry $L^\infty(\mu_{\text{sf}}) \rightarrow (L^1(\mu))^*$ is onto.
3. The multiplication image on $L^2(\mu)$ is a von Neumann algebra.
4. That image is a maximal abelian $*$ -subalgebra of $B(L^2(\mu))$.

This is the reduced-measure form of the standard duality and representation results [1, 2]. Order completeness refers to the real/self-adjoint lattice, or equivalently to its projection Boolean algebra.

The dual norm. The map sends $[f]$ to $g \mapsto \int fg d\mu$. Its norm is at most $\|f\|_{\infty, \text{loc}}$ because an L^1 function is supported on a countable union of finite- μ sets. For $0 \leq c < \|f\|_{\infty, \text{loc}}$ choose a finite positive-measure $A \subseteq \{|f| > c\}$ and use

$$g_A = \frac{\chi_A}{\mu(A)} \frac{\overline{f}}{|f|}.$$

The phase is zero where $f = 0$. Then $\|g_A\|_1 = 1$ and $\int fg_A d\mu > c$. This proves isometry; it does not assert that a single test attains the supremum.

3. The qualified Radon–Nikodym face

Put $\lambda = \mu_{\text{sf}}$. A positive measure ν on Σ is *strongly λ -semifinite* when

$$\nu(E) > 0 \implies \exists F \in \Sigma : F \subseteq E, \quad \lambda(F) < \infty, \quad 0 < \nu(F) < \infty.$$

It is *λ -semifinite* if the same condition is required without the upper bound $\nu(F) < \infty$. Strong λ -semifiniteness is equivalent to λ -semifiniteness together with semifiniteness of ν .

A measure ν has *λ -support* if there is $S \in \Sigma$ such that $\nu(E) = \nu(E \cap S)$ for all E , and $\nu(F) > 0$ for every measurable $F \subseteq S$ with $\lambda(F) > 0$.

Density theorem [2, Theorem 4.4]. If λ is localizable, the following are equivalent for a positive measure ν :

1. There is measurable $h : X \rightarrow [0, \infty)$ such that

$$\nu(E) = \int_E h \, d\lambda \quad (E \in \Sigma).$$

2. $\nu \ll \lambda$ and ν is strongly λ -semifinite.
3. $\nu \ll \lambda$, ν is semifinite, and ν has λ -support.

The density is unique λ -almost everywhere. Here the range $[0, \infty)$ requires finite values. Extended-valued density theorems have different hypotheses; see [2, Corollary 4.5].

Finite ν : retain the finite-piece condition. When $\nu(X) < \infty$, the condition on $\nu(F)$ being finite becomes automatic. The remaining condition is

$$\nu \ll \lambda \quad \text{and} \quad \nu(E) > 0 \implies \exists F \subseteq E : \lambda(F) < \infty, \quad \nu(F) > 0.$$

The density then lies in $L^1(\lambda)_+$. Absolute continuity is exactly vanishing on locally μ -null sets, but does not replace the displayed finite-piece condition in the unrestricted localizable theorem. Removing it in full generality introduces the set-theoretic issue discussed in [2, Remark 4.8(3)]. This restores the existing project v1.7 wording [3].

Equivalent fifth structural condition. Since λ is semifinite, localizability is equivalent to the statement that *every* positive $\nu \ll \lambda$ that is strongly λ -semifinite admits the above finite-valued density [2, Theorem 5.2(vi)]. A finite-measure corollary alone is not this entire quantified property.

Scope. This statement concerns μ_{sf} . Adding a new observation to a representation leaves μ unchanged and does not retroactively make the old representation faithful. Related constructions that change the underlying space belong to a separate completion problem; Bouafia–De Pauw [4] is a reference for that direction, not an extra hypothesis used here.

4. Examples A and B: visibility and size

A. An infinite-mass point

Let $X = [0, 1] \sqcup \{p\}$, with the usual Lebesgue measurable sets on the interval, $\mu = \lambda_{\text{Leb}}$ there, and $\mu(\{p\}) = \infty$. Every finite- μ set excludes p . Thus

$$L^2(\mu) \cong L^2[0, 1], \quad L^1(\mu) \cong L^1[0, 1].$$

The reduction is Lebesgue measure on the interval and zero at p . It is localizable, although $\mu \neq \mu_{\text{sf}}$. With $f = \chi_{\{p\}}$,

$$\operatorname{ess\,sup}_{\mu} |f| = 1, \quad \|M_f\| = \|f\|_{\infty, \text{loc}} = 0.$$

The set $\{p\}$ is locally null but not μ -null. This is an exact counterexample to unqualified faithfulness and the raw essential-supremum norm identity.

The measure $\nu = \infty \delta_p$ fails absolute continuity with respect to μ_{sf} and fails strong μ_{sf} -semifiniteness. No measurable nonnegative density, finite or extended-valued, can represent it against μ_{sf} : every integral over $\{p\}$ is zero.

To restore this particular distinction, use the enlarged Hilbert representation

$$\rho(f) = M_{f|_{[0,1]}} \oplus f(p)I_{\mathbb{C}} \quad \text{on } L^2[0, 1] \oplus \mathbb{C}.$$

Its image is $L^\infty[0, 1] \oplus \mathbb{C}$. The additional evaluation detects p while preserving the old multiplication coordinate. Ordinary interval-null differences remain invisible. The measure itself has not changed.

B. An uncountable disjoint union

Let I be uncountable and let $X = \bigsqcup_{i \in I} [0, 1]_i$ with the *direct-sum sigma-algebra*: a set is measurable exactly when its intersection with each slice is Lebesgue measurable. Define

$$\mu(E) = \sum_{i \in I} \lambda_i(E \cap [0, 1]_i),$$

where a nonnegative uncountable sum is the supremum of its finite subsums. This space is strictly localizable and semifinite, but not σ -finite. One has

$$L^2(\mu) = \bigoplus_{i \in I} L^2[0, 1], \quad (L^1(\mu))^* \cong L^\infty(\mu).$$

The full multiplication algebra is maximal abelian and weakly closed. Non- σ -finiteness alone causes none of the structural failures.

If a finite $\nu \ll \mu$ also satisfies the finite-piece condition, it is concentrated on countably many slices of positive ν -mass. Indeed only countably many slices can have positive mass; a positive remainder would contain a finite- μ subset of positive ν -mass, contradicting the zero mass of its countably many positive- μ slices and absolute continuity. Ordinary σ -finite RN then supplies an L^1 density. Do not omit the finite-piece assumption in this argument.

5. Example C: missing assemblies

For uncountable I , let

$$\Sigma = \{A \subseteq I : A \text{ is countable or } I \setminus A \text{ is countable}\},$$

and let μ be counting measure restricted to Σ . Its only null set is $\emptyset$. Each singleton has measure one, so μ is semifinite and $\mu = \mu_{\text{sf}}$.

Choose $J \subset I$ such that J and $I \setminus J$ are uncountable. The measurable family $\{\{j\} : j \in J\}$ has no supremum in Σ . Any measurable upper bound H is co-countable. Since $H \neq J$, remove some $k \in H \setminus J$ to obtain a strictly smaller measurable upper bound. Hence the measure algebra is not Dedekind complete.

Every summable function has countable support, and such functions are measurable. Therefore

$$L^1(\mu) = \ell^1(I), \quad L^2(\mu) = \ell^2(I), \quad (L^1(\mu))^* = \ell^\infty(I).$$

The original $L^\infty(\mu)$ consists only of bounded Σ -measurable functions, a proper subset of $\ell^\infty(I)$. In particular χ_J defines a bounded functional on $\ell^1(I)$ but has no measurable representative in the original algebra. The canonical inclusion is isometric and injective, but not onto.

RN failure even with the finite-piece condition. Define

$$\nu_J(A) = \#(A \cap J), \quad A \in \Sigma,$$

with infinite counts assigned ∞ . This is an absolutely continuous, strongly μ -semifinite measure: any positive set contains a singleton from J . A density would have to equal χ_J by testing every singleton, which is impossible because $J \notin \Sigma$. The missing assumption here is localizability.

A finite counterexample. Define $\eta(A) = 0$ for countable A and $\eta(A) = 1$ for co-countable A . Then η is finite and absolutely continuous with respect to μ , but has no density. Singleton tests would force a density to vanish everywhere. It also fails μ -semifiniteness. Thus “ ν is Σ -measurable” is not an adequate RN criterion: these measures are already defined on Σ .

Scoreboard.

Property	A	B	C
$\mu = \mu_{\text{sf}}$	No	Yes	Yes
μ_{sf} localizable	Yes	Yes	No
μ is σ -finite	No	No	No
Raw essential-supremum norm holds	No	Yes	Yes
Canonical $L^\infty(\mu_{\text{sf}}) \rightarrow (L^1)^*$ onto	Yes	Yes	No
Qualified RN property for all eligible ν	Yes	Yes	No

The failures in A and C require different repairs. B shows why σ -finiteness is stronger than the structural theorem needs.

6. Exact witness recovery

For this translation, declare a source domain D , a total retained map $\pi : D \rightarrow Q$ with $Q = \pi(D)$, and a witness W . Define

$$N_W(\pi) = \{q \in Q : \exists x, y \in \pi^{-1}(q), W(x) \neq W(y)\}.$$

The witness descends to Q exactly when this set is empty. The canonical exact refinement retaining both the old label and the witness is

$$R = \{(\pi(x), W(x)) : x \in D\}, \quad (q, w) \mapsto q.$$

If another retained record r determines both π and W , it maps to R . This is the sense in which R is coarsest. It supplies no dynamics, topology, or global shape by itself.

A. The function encoding works

Use $D = \mathcal{B}(X, \Sigma)$ from example A, $\pi(f) = M_f$, and $W(f) = f(p)$. For every f ,

$$\pi(f + \chi_{\{p\}}) = \pi(f), \quad W(f + \chi_{\{p\}}) = W(f) + 1.$$

Thus $N_W(\pi) = Q$, including fibers whose chosen representative has $f(p) = 0$. The refinement has image

$$f \mapsto (M_{f|_{[0,1]}}, f(p)) \in L^\infty[0, 1] \oplus \mathbb{C}.$$

The witness $W(f) = f(p)$ stays fixed; the retained map is enlarged from π to (π, W) . In operator language, the representation changes from π to ρ . Neither change alters μ .

Why the point encoding does not isolate p . If $D = X$ and $\pi(x) = [\{x\}]$ in $\Sigma/\mathcal{N}_{\text{loc}}$, then *every* point maps to zero. Each interval singleton is Lebesgue-null and $\{p\}$ is locally null. With $W(x) = x$, $Q = \{0\}$ and $N_W(\pi) = \{0\}$. This encoding loses all point identities. It cannot describe interval singletons as visible while singling out p as invisible.

B. Full and restricted observations

Use the full direct-sum multiplication map and $W(f) = [f]_\mu$. Semifiniteness gives $N_W(\pi) = \emptyset$: the operator determines the almost-everywhere class.

For a countable $K \subset I$, instead retain $\pi_K(f) = M_{f|_{X_K}}$, where $X_K = \bigsqcup_{i \in K} [0, 1]_i$. Keep the same full-field witness W . Altering f by a constant on any omitted slice preserves $\pi_K(f)$ and changes $W(f)$. Every attained label therefore lies in $N_W(\pi_K)$. This is a static information-loss statement; a countable set of spatial coordinates is not a temporal history.

7. Completeness is a different obstruction

In example C, set

$$A_\Sigma = \{f : I \rightarrow \mathbb{C} : f \text{ bounded and } \Sigma\text{-measurable}\}.$$

The legitimate map is the inclusion

$$\iota : A_\Sigma \hookrightarrow \ell^\infty(I) = (\ell^1(I))^*.$$

It is one-to-one. For $D = A_\Sigma$, $\pi = \iota$, and $W(f) = f$, every attained fiber is a singleton, so $N_W(\pi) = \emptyset$. Yet ι is not onto.

The missing object lies outside the image. For J as in example C,

$$\Phi_J(g) = \sum_{j \in J} g(j), \quad g \in \ell^1(I),$$

is a bounded functional represented by $\chi_J \in \ell^\infty(I)$. It lies outside $\iota(A_\Sigma)$. The set

$$(\ell^1(I))^* \setminus \iota(A_\Sigma)$$

is a representability gap, not the non-descent locus defined on attained labels. Calling an operation “take a measurable representative when one exists” does not define a total map on all of $\ell^\infty(I)$.

There is no canonical inner core or outer hull in Σ . An uncountable J with uncountable complement has no greatest countable subset and no least co-countable measurable superset. Choosing arbitrary bounds would add an arbitrary observation rule. Recording all measurable lower bounds would instead distinguish every subset, since the family includes every singleton it contains. Neither construction yields the claimed collision map without further specification.

A precise completion witness. For finite $F \subset J$, the projections M_{χ_F} belong to the original multiplication algebra. Directed by inclusion of finite F , they converge strongly on $\ell^2(I)$ to M_{χ_J} :

$$\|(M_{\chi_J} - M_{\chi_F})g\|_2^2 = \sum_{j \in J \setminus F} |g(j)|^2 \longrightarrow 0.$$

The limit does not belong to the original algebra. This explicitly proves failure of strong closure. It also disproves maximal abelianness, since the omitted diagonal operator commutes with every original multiplier.

Adjoining all subsets supplies the Boolean completion $\mathcal{P}(I)$, with counting measure; its bounded-function algebra is $\ell^\infty(I)$. This realizes the missing functionals in this example. It expands what can be represented, while the inclusion above was already faithful on its original domain.

Practical distinction. A same-label collision asks whether one observed label determines a witness. A completeness question asks whether the allowed algebra can express an object assembled from available pieces. Neither question can stand in for the other.

8. A fully declared dynamics example

The original investigation made one-step prediction claims from static slice restrictions. Supply dynamics before assigning a prediction diameter.

Use example B with states $D = L^\infty(\mu)$, so state equality means almost-everywhere equality. Fix nonempty countable $K \subset I$, a retained slice $i_0 \in K$, and an omitted slice $k \notin K$. Let $r(f) = f|_{X_K}$. Outputs carry sigma-algebras that separate distinct slice classes, for example the product of the norm-Borel sigma-algebras on the slice L^∞ spaces.

For deterministic $U : D \rightarrow D$, use one application of U per tick and define

$$\delta_1^B(z) = \sup_{r(f)=r(g)=z} \text{TV}(\delta_{B(Uf)}, \delta_{B(Ug)}),$$

where $\text{TV}(P, Q) = \sup_E |P(E) - Q(E)|$. Distinct deterministic outputs have distance one.

Identity dynamics. Set $U(f) = f$. For output $B(f) = f_{i_0}$, $\delta_1^B(z) = 0$ on every retained fiber. For full-field output $B(f) = f$, $\delta_1^B(z) = 1$, since an omitted slice can vary. Nevertheless r itself *is* autonomous: $r(Uf) = r(f)$. Failure to recover the full field does not imply failure of the retained state's own dynamics.

Swap dynamics. Instead let U swap the complete slice functions at i_0 and k using their common $[0, 1]$ coordinate, fixing all other slices. Two states can agree on X_K and differ at k . Their next retained i_0 outputs then differ, so r is not autonomous and $\delta_1^{B_{i_0}}(z) = 1$ on every fiber. Retaining slice k repairs closure for this swap because the enlarged set of retained slices is invariant under it.

These are new, explicitly declared illustrative dynamics. They do not describe the toroidal kernel and do not prove anything about finite temporal suffixes for that kernel.

General finite horizons. Once the kernel, clock, output B , retained record r , and horizon h are fixed, compare laws of $(B_1, \dots, B_h)$ over each r -fiber. Longer horizons cannot reduce the diameter. Refinement restricts the pairs compared and cannot enlarge a refined fiber's diameter. With $B = r$, zero one-step diameter on all fibers is the relevant universal closure condition. A single tested pair only supplies a lower bound.

Corrected dictionary.

Question	Appropriate construction
Same label, different witness?	Total map π , attained image Q , and $N_W(\pi)$.
Recover both label and witness?	Image of (π, W) , projecting back to Q .
Can the algebra express the needed assembly?	Suprema, completion, or surjectivity of a specified inclusion.
Does the retained record predict its update?	Declared dynamics and equality of next retained-output laws.

9. Repairs, provenance, and scope

Repairs to the supplied investigation.

1. Restored the finite- ν finite-piece condition from project v1.7; supplied the missing definition of λ -support. Qualified the slice-concentration argument and replaced the misleading RN scoreboard row.
2. Kept the infinite-point function example. Used $f + \chi_{\{p\}}$ to exhibit a collision for every f , including $f(p) = 0$. Corrected the point encoding: all singleton labels are zero there.
3. Kept W fixed when refining π to (π, W) . Specified the enlarged Hilbert representation separately.
4. Replaced the undefined subset-bound map and partial measurable-representative map with an injective inclusion, a concrete omitted functional, and a strong-limit witness of incompleteness.
5. Added explicit identity and swap dynamics before computing δ_1 . Distinguished full-field recovery from autonomy of the retained state and spatial restriction from temporal memory.
6. Located the actual project v1.7 in Drive [3]. Its absence from a separately located public repository file is not absence from the project sources.

The earlier core repairs remain: separate reduction from completion; declare the L^2 witness; avoid treating abstract von Neumann algebra uniqueness as equivalence of arbitrary concrete representations; and normalize the complex-phase L^1 test. Different representations can have different multiplicities and commutants.

The unrelated numerical appendix also retains its correction:

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad (\sqrt{2})^{\sqrt{3}} \approx 1.822634654966, \quad \varphi(\sqrt{2})^{\sqrt{3}} \approx 2.949084820809.$$

The four-decimal value is 2.9491. Gelfond–Schneider gives transcendence of $(\sqrt{2})^{\sqrt{3}}$, hence of its nonzero algebraic multiple. This number plays no role in the measure-theory argument.

Source map.

- [1] D. H. Fremlin, *Measure Theory*, Chapter 24, 243G: canonical L^∞ – $(L^1)^*$ injectivity and surjectivity. The construction is in the classical Segal/Fremlin lineage.
- [2] D. P. Blecher, S. Goldstein, and L. E. Labuschagne, *Abelian von Neumann algebras, measure algebras and L^∞ -spaces*: Theorem 4.4, Remark 4.8(3), and Theorem 5.2.
- [3] Project *Hidden Quotient - Core, Witness, Strike, and Retained Context*, v1.7, Part I §8 and Bridge II; and *Geometry + GQG Working Master*, §§1–2, 6.
- [4] P. Bouafia and T. De Pauw, *Radon-Nikodýmification of arbitrary measure spaces*, Extracta Mathematicae 38(2) (2023), 193–203; author’s publication listing. Related reading on changing the space; no additional construction theorem from it is invoked here.

Standing. The measure theory, the declared geometry examples, and empirical applications retain separate claims. Seat 2 and Seat 5 remain declared readings. Retained-context, Open-Seam, constraint-demotion, simplex, percentile, and lifecycle instruments are companion constructions. This correction supplies no new empirical result, global shape, holonomy, physical mechanism, social attribution, or change to the chemistry freeze or physical Q2 status.